



Republic of the Philippines
PAMANTASAN NG LUNGSOD NG MAYNILA

University of the City of Manila

General Luna, corner Muralla St, Intramuros, Manila, 1002 Metro Manila



Spelling Game

Under the Supervision of

Engr. Elsa S. Pascual

Computer Architecture and Organization (CSC 0222)

Submitted by:

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Cacal, Charles R.
Calderon, Jonas V.
Encontad, Ellysa Kateann M.
Lumba, Shania Janelle R.
Martin, Scheanz Yonce P.
Ochoa, Micah Gabriel B.
Villarico, Trazy A.



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I. MEMBERS AND TASKS

Name	Task/s
Acupan, Raphael Gio B.	• Group Leader • Game Programmer
Bisnan, Leinard Allen B.	• Sprite Artist • Design and Labor
Biblanias, Alliah Jade B.	• Sprite and Background Artist • Brochure and Presentation preparation
Cacal, Charles R.	• Game Programmer • Sprite and Background Artist • Documentation
Calderon, Jonas V.	• Sprite Artist • Design and Labor
Encontad, Ellysa Kateann M.	• Sprite and Background Artist • Brochure and Presentation preparation
Lumba, Shania Janelle R.	• Sprite and Background Artist • Dance Coordinator



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	• Booth Design
Martin, Scheanz Yonce P.	• Sprite and Background Artist • Design and Labor
Ochoa, Micah Gabriel B.	• Sprite Artist • Design and Labor
Villarico, Trazy A.	• Sprite and Background Artist • Dance Coordinator • Booth Design



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II. INTRODUCTION

This project is a spelling game developed as part of the Computer Architecture and Organization course. The game is designed primarily for toddlers and early childhood learners, offering a fun and interactive way to practice spelling through visual cues and timed challenges. To make the experience more engaging and familiar for young players, the game adopts a Toy Story-inspired visual theme, incorporating characters and design elements from the franchise as decorative and motivational elements throughout the interface.

The main goal of the game is simple: players are shown a picture of an object alongside a scrambled version of its name, and they must type the correct spelling, and spelling it quicker will yield more points. This approach to learning encourages children to connect visual representations with written words, supporting both vocabulary development and early reading skills. The addition of a timing multiplier per guess introduces a mild sense of urgency that keeps players attentive and engaged throughout the session.

The game supports two players, allowing children to take turns and compete against each other in a friendly, structured format. Each player faces the same set of ten spelling challenges, and their scores are tracked and displayed at the end of the game on a leaderboard that ranks them by the number of correct answers. This competitive element adds replayability and motivates players to improve their performance across multiple sessions.

Three difficulty levels are available: Easy, Normal, and Hard. Each adjusting the length of the word to guess. This makes the game accessible to learners at different stages of development, from beginners who need more time to think to more advanced players seeking a greater challenge. Overall,



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this project aims to demonstrate how interactive software can be used as an effective and enjoyable tool for early childhood education.



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III. DEFINITION OF TERMS

- **Correct Answer** – A player's response that exactly matches the spelling of the target word displayed during a round. A correct answer adds one point to the player's score.
- **Difficulty Level** – A setting selected at the start of the game that determines how much time each player is given to answer per round. The three available levels are Easy, Normal, and Hard, with Easy providing the most time and Hard providing the least.
- **DOSBox** – A program that allows older DOS-based software to run on modern computers. The spelling game runs inside DOSBox since it is built for a DOS environment.
- **Early Childhood Learner** – The intended audience of the game — children in the toddler to early primary stage of development, typically between the ages of two and seven, who are beginning to recognize letters and form words.
- **Game Loop** – The repeating cycle of the game in which each player goes through all ten spelling challenges one after another. The loop runs for Player 1 first, then repeats for Player 2 with the same set of words.
- **Game Over Screen** – The final screen displayed after both players have completed all ten rounds. It shows the leaderboard with each player's rank and total score, and offers the option to play again or exit the game.
- **Incorrect Answer** – A player's response that does not match the target word, or no response submitted before the timer expires. An incorrect answer does not add to the player's score.
- **Input Field** – The area on the screen where the player types their answer during a round. The player's keystrokes appear here as they type, and pressing Enter submits the answer.



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- **Leaderboard** – A rankings screen shown at the end of the game that lists both players' names and scores in order from highest to lowest. The player with more correct answers is displayed in first place.
 - **Player Name** – The display name entered by each player during the registration phase at the start of the game. Player names appear on the leaderboard at the end of the session.
 - **Player Registration** – The setup phase at the beginning of the game where each player types in their name before gameplay begins. Names are stored and used throughout the session for score tracking and leaderboard display.
 - **Round** – A single spelling challenge within the game. Each round presents the player with a sprite and a scrambled clue, starts the countdown timer, and awaits the player's typed answer. The game consists of ten rounds per player.
 - **Score** – The total number of correct answers a player accumulates across all ten rounds. Each correct answer is worth one point, for a maximum possible score of ten.
 - **Scrambled Clue** – A hint provided to the player during each round, showing the letters of the target word rearranged in a random order (for example, 'gdo' as a clue for 'dog'). The clue is displayed alongside the sprite to help the player identify the word.
 - **Spelling Game** – The interactive software application developed by the group. Players are challenged to correctly spell words associated with images displayed on the screen, within a set time limit, across ten consecutive rounds.
 - **Sprite** – A visual image displayed on the screen during each round, drawn using text characters to represent the object associated with the target word. The sprite serves as the main visual hint to guide the player toward the correct answer.



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- **Target Word** – The correct spelling of the word the player must identify and type during a given round. The target word corresponds to the object shown in the sprite and is what the scrambled clue is based on.
 - **Theme** – The visual and aesthetic style applied throughout the game's interface. This game uses a Toy Story-inspired theme, incorporating familiar characters and design elements to create an engaging and recognizable environment for young players.
 - **Two-Player Mode** – The multiplayer format of the game in which two players take turns completing the same set of ten spelling challenges. Both players' scores are tracked separately and compared on the final leaderboard.



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IV. HOW TO PLAY

The game is straightforward and easy to follow. Below is a step-by-step walkthrough of the entire game experience from start to finish.

Starting the Game

1. Launch the game through DOSBox. The Main Menu will appear on the screen, displaying the game title. Press any key to continue.

Logging In

2. The game will prompt the Player to enter a valid username and password.

Selecting player mode

4. The game will prompt the player to play either singleplayer or multiplayer.

NOTE: *If the player chooses singleplayer they will proceed to **Playing a Round (Singleplayer)**. However, if the player chooses multiplayer they will proceed to **Playing a Round (Multiplayer)**.*

Playing a Round (Singleplayer)

5. At the start of each round, a sprite/picture will appear on the middle top side of the screen. A scrambled version of the word will be shown as a clue on the right side.
6. The player must type the correct spelling of the word shown in the sprite and press Enter. The quicker they type the correct answer, the higher the points.



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7. If the answer is correct, one point is added to the player's score. If the answer is wrong no point is awarded. The game then moves on to the next round automatically.

Playing a Round (Multiplayer)

8. At the start of each round, a sprite/picture will appear on the middle top side of the screen. A scrambled version of the word will be shown as a clue on the right side.
9. A countdown timer will begin immediately. Player 1 would be the first and must type the correct spelling of the word shown in the sprite and press Enter.
10. If the answer is correct, points are added to the player's score based on how quick he was able to answer it. If the answer is wrong, no point is awarded.
11. After Player 1 is done answering in the round, it will be Player 2's turn. The mechanic would be the same as how it is for player 1. After both players are done with their turn it will now proceed to the next round.

End of Game & Leaderboard (Singleplayer)

12. After both players finish their ten rounds, the game will end and the screen will proceed to show the leaderboard (Top 5 scores).

End of Game & Leaderboard (Multiplayer)

12. After both players finish their ten rounds, the game will end and the screen will proceed to show both player's final score and the leaderboard (Top 5 scores).



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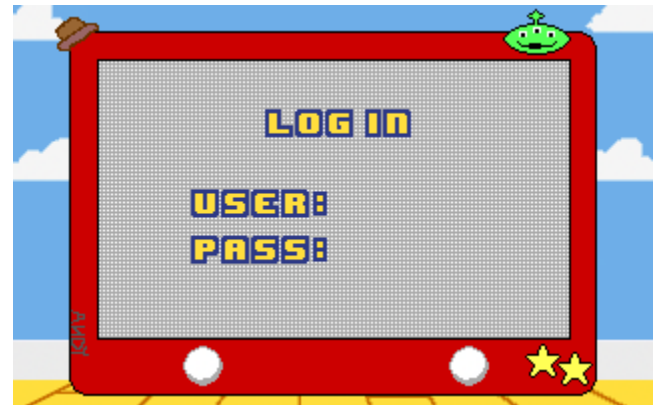
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V. STORYBOARD / UI DESIGN



START SCREEN



LOGIN SCREEN



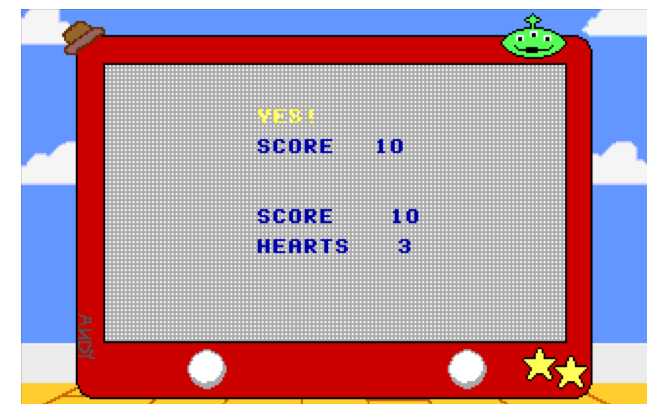
MODE SCREEN



DIFFICULTY SCREEN



GAME (P1)
SCREEN



SCORE (P1)
SCREEN



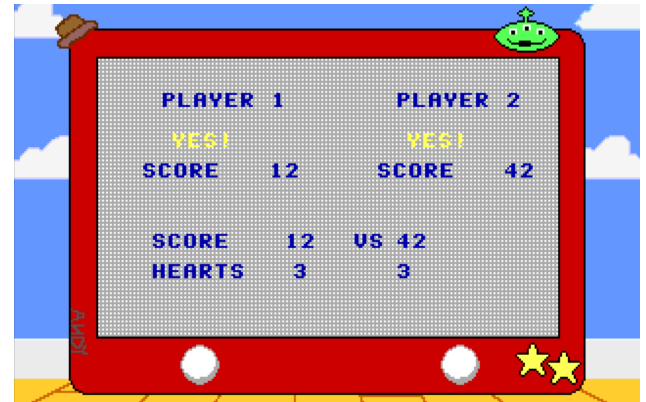
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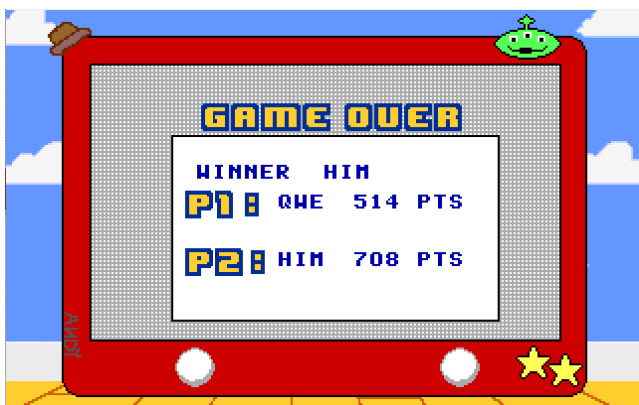
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GAME (P2)
SCREEN



SCORE (P2)
SCREEN



GAME OVER
SCREEN



LEADERBOARD
SCREEN



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VI. HOW TO INSTALL & CONFIGURE

The following is a step-by-step guide for setting up and running the game. No programming knowledge is required to follow these instructions.

Updated Source Code

- The source code can be found on:
<https://github.com/GioAcupan/spelling-game-assembly.git>

Requirements

- DOSBox 0.74 or later. downloadable at <https://www.dosbox.com>
- The game files provided by the development team (SPELL.EXE and any supporting files). This can be found on the indicated github repository link above.

Note: For the installer's convenience, the TASM and other related dependencies can be found on the repository folder [dependencies/](#).

Installation Steps

1. Download and install DOSBox on your computer if it is not already installed.
2. Open the terminal.
3. If you are inside your User folder, run the following commands:

```
cd ../../  
git clone https://github.com/GioAcupan/spelling-game-assembly.git
```
4. Open DOSBox. At the prompt, type the following commands one at a time, pressing Enter after each:

```
MOUNT C C:\spelling-game
```



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C:

PATH %PATH%; C:/DEPEND~1

5. Once you are inside the game folder, type the following to launch the game:

cd bin

SPELL

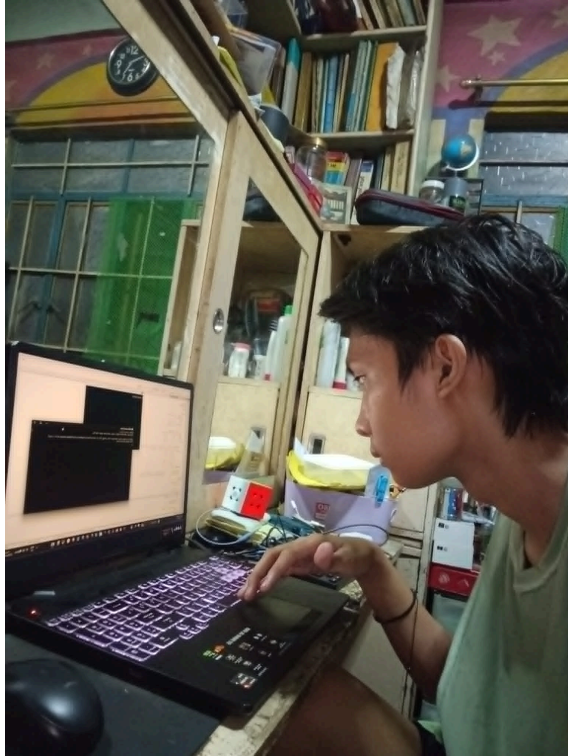
- 6 .The game will start and the Main Menu will appear. Follow the on-screen instructions to begin playing.

To exit the game at any time, close the DOSBox window directly.



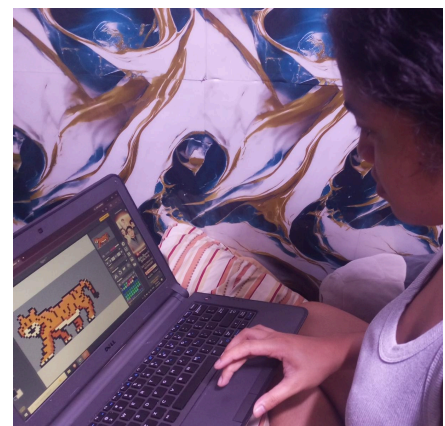
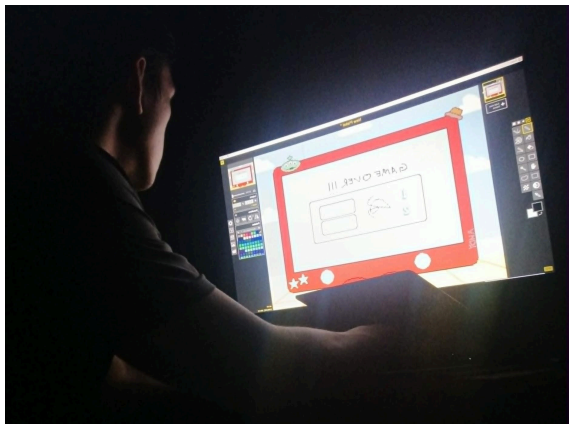
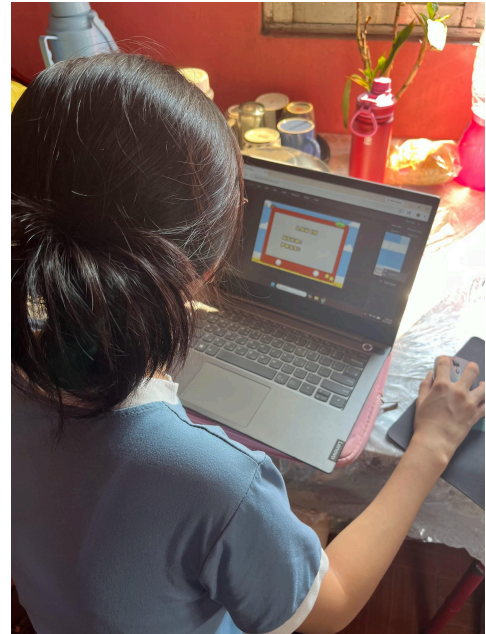
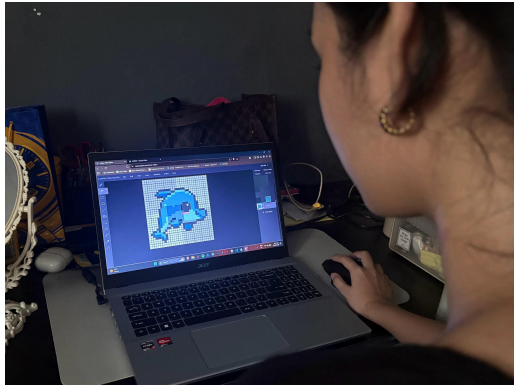
VII. Pictures During Development

Programming the Game





Sprite and Background Creation





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VIII. REFERENCES / CREDITS

Software & Tools

DOSBox Development Team. (2024). DOSBox Documentation. Retrieved from <https://www.dosbox.com/wiki/>

Microsoft Corporation. (1988). MS-DOS Programmer's Reference Manual (Version 3.3). Microsoft Press.

Thematic Reference

Pixar Animation Studios / Walt Disney Pictures. (1995–2019). Toy Story Franchise [Motion Picture Series]. Walt Disney Studios Motion Pictures. [Visual theme referenced for non-commercial, educational use only.]